# ME 210: The Joy of Curling

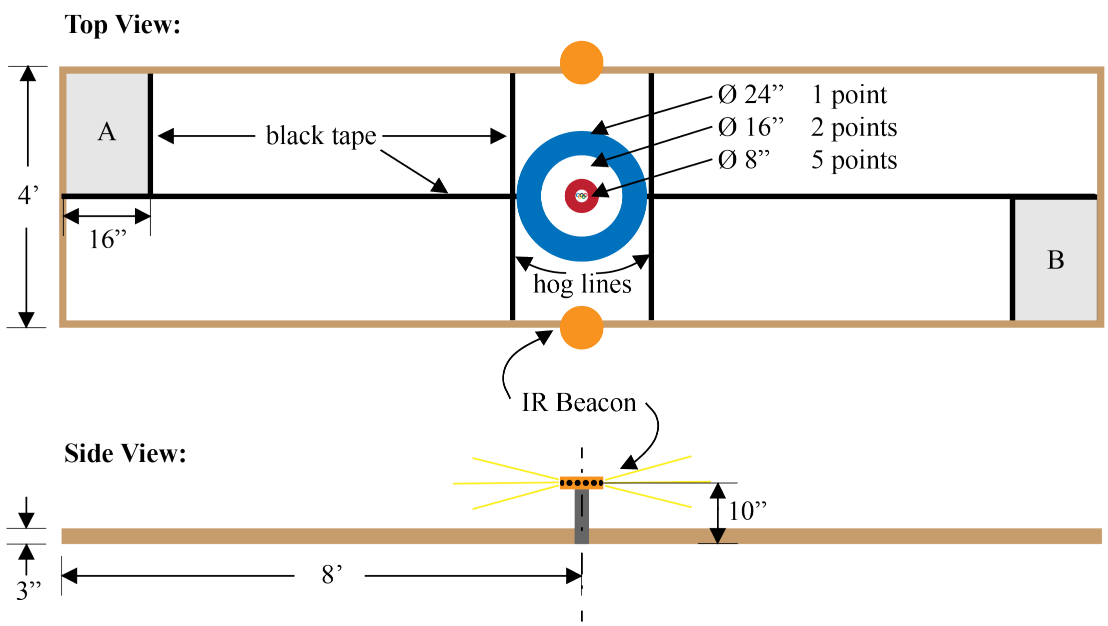
2026 Winter Olympics BOTspiel Event

**Project Presentation on March 8 starting at 6:00 pm.**

**Revision 1: 2/18/26**

## Purpose

The purpose of this project is to provide an opportunity to apply all that you have learned to solve an open-ended mechatronics design problem. The task is to design an autonomous machine that will compete against an opponent in a head-to-head version of curling, the Olympic "sport" that almost certainly ranks among the world's all-time favorites.




## Background Information

Curling — a.k.a. "Pride of the Winter Olympics", "This really stretches the word 'sport"', and "I can't believe this is an Olympic event" — originated in Scotland in the 16th century. Its proud history and rich traditions received a major shot in the arm in 1998, when, for the first time, it was included in the Winter Olympics as a regular Olympic sport. In general, the Scots can be counted upon to create bizarre, arcane contests with materials that are generally close at hand. Consider, for a moment, golf, the caber toss, the stone throw, and ferret-legging.

In the case of curling, the International Olympic Committee, after a period of careful reflection (plus a bribe, kickback, trip, and/or scholarship "gift"), has decided that simply shoving 42 lb. chunks of granite down a patch of ice at a target and then furiously scuffing or polishing the ice in its path is quaint but lacks the flash and mass market appeal that today's sporting extravaganzas require. They have cooked up what they think will be a game that can't fail to attract a large audience in a demographic sweet spot.

While the overall goal of the game remains as before (slide massive pucks across ice at a target and get them to stop as close to the center of a target's bull's-eye – the button– as possible), the competition will be enhanced by having two players play head-to-head and take their best shots simultaneously, and autonomous DROIDs (Dynamic Rolling Olympian in Disguise) will replace the human participants.

## Specifications

Each DROID must be a stand-alone entity capable of meeting all specifications while connected only to supplied battery pack(s). Your DROID will start in a randomly selected starting zone (A or B). Here, you manually load up your DROID with 3 pucks and press start, from whence the DROID must behave autonomously. DROID players will be allowed to navigate the arena (CURLING SHEET) to take shots from anywhere, as long as the DROID's starting dimensions do not pass the hog line. After playing the three pucks, the DROID must return fully into the starting zone in order to receive more pucks. A maximum of eight pucks may be played. But beware! Your opponents are also racing to place their pucks as close to the button as possible and may knock your pucks out of the way.

### Arena

The BOTspiel Winter Olympics Curling Arena consists of two particle board sheets (4 ft × 8 ft) arranged end-to-end to form a 4 ft × 16 ft CURLING SHEET.

- The bull's-eye (House) is centered on the seam between the two particle boards
- Button (center most ring) diameter: 8 inches
- Middle ring diameter: 18 inches
- Outer ring diameter: 24 inches
- Non-reflective black tape marks the center line, hog line, and bounding edges of the starting zones.
- Starting zones: 16" by 24"
- Protective perimeter border: 3.5 inches high
- Tolerances on the dimensions of the CURLING SHEET are ±1 inch unless otherwise specified. (Double check this with our CURLING SHEET once it is built.)
- Once the CURLING SHEET is constructed, its dimensions may supersede the above tolerances.

### Infrared Beacons

Infrared beacons are positioned on either side of the CURLING SHEET, where the two particle boards meet:

- Distance: 1 inch from the side edge of the CURLING SHEET
- Height: 10 inches above surface of the CURLING SHEET
- Modulation frequency: 909 Hz
- Duty cycle: 50%

**Game Rules**

At the start of a BOTspiel match (as declared audibly):

- Each DROID will begin in a starting zone (A or B) in a randomized orientation decided by the teaching staff
- Each DROID begins with up to three pucks and can only hold up to three pucks at any given time.
- No further human interaction is allowed after the DROID has started moving, except to reload more pucks.
- In order to reload more pucks, a DROID must have completely left the starting zone and returned completely into the starting zone.
- During reloading, direct human interaction with the DROID is allowed, but only for the purpose of reloading. Any other adjustments will result in a penalty (1 point).
- Each DROID must automatically cease gameplay at the end of the match (two minutes)
- Intentional jamming of your opponent's sensing abilities is prohibited
- Intentional destruction, damage, or alteration of any part of the field or other DROIDS is expressly forbidden
- the most a puck can be above the CURLING SHEET is 1 inch, after leaving the DROID.
- Members of your team are not allowed to position themselves (that's you, the humans) in a way that will interfere with the activities of any opponent's DROID. Polite, "G-rated" heckling is permitted, of course
- The tournament bracket order will be determined by the order in which teams perform the graded check-off (see performance requirements)
- Sideline coaching is strictly prohibited; no information, besides those inherent from the game, may be passed to your DROID during the match.

**Scoring**

| **Puck Location** | **Points** |
|---|---|
| Center/ Button | 5 |
| Middle ring | 2 |
| Outer ring | 1 |
| Past opponent's hog line | -3 |

- Scores are computed as the sum of all puck values
- Puck locations are determined by the center point location of the puck
- In the event of a scoring tie, the staff will preside over the final ruling of the round
- If two pucks are on top of each other, only the puck closer to the button gets scored
- If a team's puck knocks the other team's puck out of a given scoring zone, it will gain the points associated with the given scoring zone. For example, DROID A's puck knocks DROID B's puck out from the 3 point zone into the 5 point zone, and DROID's A puck ends up in the 3 point zone. For the given two pucks, DROID A gains 6 points and DROID B has 5 points.

- During the BOTspiel, if the DROID behaves in an unexpected manner, human interaction is permitted for a total manual reset (back to the starting zone) for a 3 point penalty.
- If human interaction extends beyond reloading pucks, when the DROID is in the starting zone, there is a 1 point penalty. Your DROID may need a last minute bandage worth this penalty.
- If the front of a DROID clearly passes it's own team hog line (past the black tape by a centimeter or so), there is a 1 point penalty and all shots taken past the hog line do not count. To avoid this 1 point penalty and reach over the hog line, a DROID must have a mechanical mechanism that expands the DROID beyond its original starting zone dimensions, deployed outside of the starting zone. Only portions that expand from the DROID's original starting zone dimensions may pass the hog line.

### Puck Specifications

- Material: hardened steel chrome with ABS plastic top cover
- Diameter: 2.08 inches
- Thickness: 1 inches
- Can be slid on either the steel chrome face or colored plastic face
- Weight: 8 oz.

### DROID Constraints

- Maximum footprint in starting zone: 13 in × 13 in
- Maximum footprint out of starting zone: 20 in × 20 in
- Maximum height: 12 in
- Must contain all pucks internally (fits in the DROID footprint)
- Front of DROID, as defined by DROID dimensions in the starting zone, cannot cross the hog line by over a centimeter.
- Operates as a stand-alone entity, operating completely untethered
- Power must be supplied by the provided batteries only (two 7.2V NiMH rechargeable battery packs)
- Each DROID should incorporate an easily accessible toggle switch on its exterior, which will serve as an emergency stop switch. The switch must cut all power to the DROID when toggled.
- A fuse must be installed.
- Control software should be executed from the flash memory of one or more Arduino UNO micro-controllers. Any other micro-controller usage must be approved by Professor Tom Kenny.
- Must adhere to an expenditure limit of $200 USD for the materials and parts used in the construction of the DROID. The cost of the provided batteries, fuses/circuit breakers, and the lab kit components (including a single Arduino per team member) do not count towards this limit.
- Robust enough to deal with all normal game interactions including, but not limited to, collisions with any part of the CURLING SHEET, pucks, and debris
- Must be constructed as part of ME 210 activities during the remainder of the quarter. It may not be based on a commercial or otherwise preexisting platform. Rulings from a member of the staff may be requested if there are questions about the content of your DROID.